Exercise C - II

1. Let p and q be the two roots of the equation, $mx^2 + x(2-m) + 3 = 0$. Let m_1, m_2 be the two values of m satisfying $\frac{p}{q}+\frac{q}{p}=\frac{2}{3}$. Determine the numerical value of $\frac{m_1}{m_2^2}+\frac{m_2}{m_1^2}$.

यदि p तथा q समीकरण $mx^2 + x(2-m) + 3 = 0$ के दो मूल है। यदि m_1, m_2, m के दो मान है जो $\frac{p}{q}+\frac{q}{p}=\frac{2}{3}$ को संतुष्ट करते है। $\frac{m_1}{m_2^2}+\frac{m_2}{m_1^2}$ का आंशिक मान ज्ञात कीजिए।

Ans. 99

Sol. $mx^2 + x(2-m) + 3 = 0 \left\langle \begin{matrix} p \\ q \end{matrix} \right.$

$p+q=\frac{m-2}{m}, pq=\frac{3}{m}$

$\Rightarrow p^2 + q^2 = (p + q)^2 - 2pq$

$\Rightarrow p^2+q^2=\frac{(m-2)^2}{m^2}-\frac{6}{m}$

$=\frac{(m-2)^2-6m}{m^2}$

$=\frac{m^2+4-10m}{m^2}$

$\Rightarrow \frac{p}{q}+\frac{q}{p}=\frac{2}{3}$

$\frac{p^2+q^2}{pq}=\frac{2}{3}$

$3\cdot(p^2 + q^2) = 2pq$

$3\cdot\frac{(m^2+4-10m)}{m^2}=2\cdot\frac{3}{m}$

$m^2 + 4 - 10m = 2m$

$m^2-12m+4=0 \left\langle \begin{matrix} m_1 \\ m_2 \end{matrix} \right.$

$m_1 + m_2 = 12,\ m_1 \cdot m_2 = 4$

$\Rightarrow m_1^3+m_2^3=(m_1+m_2)^3-3m_1m_2(m_1+m_2)$

$= (12)^3 - 3 \times 4 \times 12$

$= 12^3 - 12^2$
$= 12^2 (12 - 1)$
$= 144 \times 11$

$\Rightarrow \frac{m_1}{m_2^2} + \frac{m_2}{m_1^2}$

$= \frac{m_1^3 + m_2^3}{m_1^2 \cdot m_2^2}$

$= \frac{144 \times 11}{4 \times 4}$

$= 99$

2. Solve for x the following equation : $\log_{(2x+3)}(6x^2 + 23x + 21) = 4 - \log_{(3x+7)}(4x^2 + 12x + 9)$

x के मान ज्ञात करने के लिए निम्न समीकरण को हल करें : $\log_{(2x+3)}(6x^2 + 23x + 21) = 4 - \log_{(3x+7)}(4x^2 + 12x + 9)$

Ans. $x = -\frac{1}{4}$

Sol. $\log_{(2x+3)}(6x^2 + 23x + 21) = 4 - \log_{(3x+7)}(4x^2 + 12x + 9)$
$\log_{(2x+3)} (2x + 3)\cdot(3x + 7) = 4 - \log_{(3x+7)}\cdot(2x + 3)^2$
$1 + \log_{(2x+3)}(3x + 7) = 4 - 2\log_{(3x+7)} (2x + 3)$

put $t = \log_{(2x+3)} (3x + 7)$ $\Rightarrow \frac{1}{t} = \log_{(3x+7)} (2x + 3)$

$t + \frac{2}{t} - 3 = 0$
$t^2 - 3t + 2 = 0$
$t^2 - 2t - t + 2 = 0$
$t(t - 2) - 1(t - 2) = 0$
$(t - 1) (t - 2) = 0$
$t = 1, 2$
for $t = 1$
$\log_{(2x+3)} (3x + 7) = 1$
$2x + 3 = 3x + 7$
$x = -4$
for $t = 2$
$\log_{(2x+3)} (3x + 7) = 2$
$(2x + 3)^2 = 3x + 7$
$4x^2 + 9 + 12x - 3x - 7 = 0$
$4x^2 + 9x + 2 = 0$
$4x^2 + 8x + x + 2 = 0$
$4x(x + 2) + 1 (x + 2) = 0$
$(x + 2) (4x + 1) = 0$

$x = -2, -\frac{1}{4}$

at $x = -4$

$2x + 3 = -5 < 0$ (wrong)

$3x + 7 = -5 < 0$ (wrong)

at $x = -2$

$2x + 3 = -1 < 0$ (wrong)

$3x + 7 = 1 < 0$

at $x = -\frac{1}{4}$

$2x + 3 = \frac{2}{-4} + 3 = \frac{5}{2} > 0$

$3x + 7 = 7 - \frac{3}{2} = \frac{11}{2} > 0$

$\therefore x = -\frac{1}{4}$

3. Find the values of 'a' for which $-3 < [(x^2+ax-2)/(x^2+x+1)] < 2$ is valid for all real x.

'a' का मान ज्ञात कीजिए, यदि असमिका $-3 < [(x^2+ax-2)/(x^2+x+1)] < 2$, x के वास्तविक मानों के लिए सत्य है।

Ans. $-2 < a < 1$

Sol. $-3 < \frac{x^2 + ax - 2}{x^2 + x + 1} < 2$

when		and when
$\frac{x^2 + ax - 2}{x^2 + x + 1} > -3$		$\frac{x^2 + ax - 2}{x^2 + x + 1} < 2$
$4x^2 + x\cdot(a + 3) + 1 > 0$		$x^2 + (2 - a)x + 4 > 0$
$D < 0$		$D < 0$
$(a + 3)^2 - 16 < 0$		$(a - 2)^2 - 16 < 0$
$(a + 3 + 4)\cdot(a + 3 - 4) < 0$		$(a - 2 + 4)(a - 2 - 4) < 0$
$(a + 7)(a - 1) < 0$		$(a + 2)(a - 6) < 0$
$a \in (-7, 1)$	$\cap$	$a \in (-2, 6)$

$\Rightarrow a \in (-2, 1)$ or $-2 < a < 1$

4. Find the minimum value of $\frac{\left(x + \frac{1}{x}\right)^6 - \left(x^6 + \frac{1}{x^6}\right) - 2}{\left(x + \frac{1}{x}\right)^3 + x^3 + \frac{1}{x^3}}$ for $x > 0$.

धनात्मक x के लिए $\dfrac{\left(x+\frac{1}{x}\right)^6-\left(x^6+\frac{1}{x^6}\right)-2}{\left(x+\frac{1}{x}\right)^3+x^3+\frac{1}{x^3}}$ का न्यूनतम मान बताइये।

Ans. $Y_{Min} = 6$

Sol. $$\frac{\left(x+\frac{1}{x}\right)^6-\left(x^6+\frac{1}{x^6}\right)-2}{\left(x+\frac{1}{x}\right)^3+\left(x^3+\frac{1}{x^3}\right)}$$

$$=\frac{\left(x+\frac{1}{x}\right)^6-\left\{\left(x^3\right)^2+\left(\frac{1}{x^3}\right)^2\right\}-2}{\left(x+\frac{1}{x}\right)^3+\left(x^3+\frac{1}{x^3}\right)}$$

$$=\frac{\left(x+\frac{1}{x}\right)^6-\left\{\left(x^3+\frac{1}{x^3}\right)^2-2\right\}-2}{\left(x+\frac{1}{x}\right)^3+\left(x^3+\frac{1}{x^3}\right)}$$

$$=\frac{\left(\left(x+\frac{1}{x}\right)^3\right)^2-\left(x^3+\frac{1}{x^3}\right)^2+2-2}{\left(x+\frac{1}{x}\right)^3+\left(x^3+\frac{1}{x^3}\right)}$$

$$=\frac{\left[\left(x+\frac{1}{x}\right)^3+x^3+\frac{1}{x^3}\right]\cdot\left[\left(x+\frac{1}{x}\right)^3-\left(x^3+\frac{1}{x^3}\right)\right]}{\left[\left(x+\frac{1}{x}\right)^3+\left(x^3+\frac{1}{x^3}\right)\right]}$$

$$=x^3+\frac{1}{x^3}+3\cdot\left(x+\frac{1}{x}\right)-x^3-\frac{1}{x^3}$$

$$=3\left(x+\frac{1}{x}\right)$$

$$\because x+\frac{1}{x}\geq 2$$

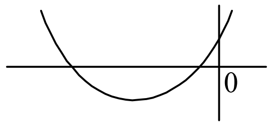
$\Rightarrow$ Miimum value $= 3 \times 2$
$= 6$

5. Find all numbers p for each of which the least value of the quadratic trinomial $4x^2 - 4px + p^2 - 2p + 2$ on the interval $0 \le x \le 2$ is equal to 3.

'p' के किन मानों के लिए द्विघात $4x^2 - 4px + p^2 - 2p + 2$ का अन्तराल $0 \le x \le 2$ में न्यूनतम मान 3 है।

Ans. $p = 1 - \sqrt{2}$ or $5 + \sqrt{10}$

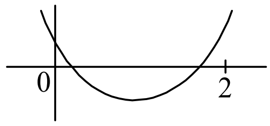
Sol. $4x^2 - 4px + p^2 - 2p + 2 = 0$
Case–I :



$-\frac{b}{2a} < 0$

$\frac{4p}{8} < 0$

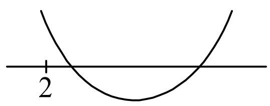
$p < 0$
$f(0) = 3$
$p^2 - 2p + 2 - 3 = 0$
$p^2 - 2p - 1 = 0$
$p = 1 + \sqrt{2}, 1 - \sqrt{2}$
$\Rightarrow p = 1 - \sqrt{2}$
Case–II :



$0 < -\frac{b}{2a} < 2$

$0 < \frac{p}{2} < 2$

$0 < p < 4$
Case–III :



$-\frac{b}{2a} > 2$

$\frac{p}{2} > 2$

$f(2) = 3$
$16 - 8p + p2 - 2p + 2 = 3$
$p^2 - 10p + 15 = 0$
$p = 5 + \sqrt{10}, 5 - \sqrt{10}$
$\therefore p = 1 - \sqrt{2}, 5 + \sqrt{10}$

6. Find the values of 'b' for which the equation $2\log_{\frac{1}{25}}(bx+28) = -\log_5(12-4x-x^2)$ has only one solution.

'b' के किन मानों के लिए समीकरण $2\log_{\frac{1}{25}}(bx+28) = -\log_5(12-4x-x^2)$ का केवल एक हल होगा।

Ans. $(-\infty,-14)\cup\{4\}\cup\left[\frac{14}{3},\infty\right)$

Sol. $2\log_{\frac{1}{25}}(bx+28) = -\log_5(12-4x-x^2)$

$-\log_5(bx+28) = -\log_5(12-4x-x^2)$

$\Rightarrow x^2 + (b+4)x + 16 = 0$

for only one solution

$D = 0$

$(b+4)^2 = 64$

$b = 4, -12$

at $b = 4$

$x^2 + 8x + 16 = 0$

$x = -4$

$\Rightarrow$ at $b = 4$, $x = -4$

$bx + 28 = 4\cdot(-4) + 28 = 12 > 0$; $12 - 4x - x^2 = 12 + 16 - 16 = 12 > 0$

at $b = -12$

$x^2 - 8x + 16 = 0$

$x = 4$

at $b = -12$, $x = 4$

$bx + 28 = -48 + 28 = -20 < 0$ (rejected)

$\Rightarrow bx + 28 > 0$ And $12 - 4x - x^2 > 0$

$x > \frac{-28}{b}$ $\qquad x^2 + 4x - 12 < 0$

$x > \frac{-28}{4}$ $\qquad (x+6)(x-2) < 0$

$x > -7$ $\qquad x \in (-6, 2)$

Itersection value

$x \in (-6, 2) \leftarrow$ domain of log

$f(-6)\cdot f(2) < 0$

$(28 - 6b)\cdot(28 + 2b) < 0$

$b \in (-\infty,-14)\cup\left(\frac{14}{3},\infty\right)$

at $b = -14$

$x = 8, 2$ (not lies in domain)

at $b = \frac{14}{3}$ (x lies in domain)

$$\Rightarrow b \in (-\infty, -14) \cup \{4\} \cup \left[\frac{14}{3}, \infty\right)$$

7. Solve the system of equations :

$\log_a x \ \log_a(xyz) = 48$, $\log_a y \ \log_a(xyz) = 12$, $\log_a z \ \log_a(xyz) = 84$ where $a > 0$, $a \neq 1$

समीकरणों के निकाय को हल करें :

$\log_a x \ \log_a(xyz) = 48$, $\log_a y \ \log_a(xyz) = 12$, $\log_a z \ \log_a(xyz) = 84$ जहाँ $a > 0$, $a \neq 1$

Ans. (a^4, a, a^7) or $\left(\frac{1}{a^4}, \frac{1}{a}, \frac{1}{a^7}\right)$

Sol. $\log_a x \cdot \log_a(xyz) = 48$ ...(1)

$\log_a y \cdot \log_a(xyz) = 12$ ...(2)

$\log_a z \cdot \log_a(xyz) = 84$ ...(3)

By (3) ÷ (1)

$$\log_x z = \frac{7}{4}$$

$$z = x^{\frac{7}{4}}$$

$$x = z^{\frac{4}{7}}$$

By (3) ÷ (2)

$$\log_y z = \frac{84}{12}$$

$$z = y^7$$

$$y = z^{\frac{1}{7}}$$

from equation (1)

$\log_a x \cdot \log_a(xyz) = 48$

$$\log_a z^{\frac{4}{7}} \cdot \log_a\left(z^{\frac{4}{7}} \cdot z^{\frac{1}{7}} \cdot z\right) = 48$$

$$\frac{4}{7} \cdot \frac{12}{7} \cdot (\log_a z)^2 = 48$$

$(\log_a z)^2 = (7)^2$

$\log_a z = \pm 7$

$z = a^7, a^{-7}$

$$\Rightarrow x = \left(a^{\frac{4}{7}}\right)^7 = a^4 \text{ or } a^{-4}$$

$$y = z^{\frac{1}{7}} = \left(a^7\right)^{\frac{1}{7}} = a \text{ or } a^{-1}$$

$z = a^7$ or a^{-7}

(a^4, a, a^7) or (a^{-4}, a^{-1}, a^{-7})

8. Let $y = \sqrt{\log_2 3 \cdot \log_2 12 \cdot \log_2 48 \cdot \log_2 192 + 16} - \log_2 12 \cdot \log_2 48 + 10$. Find $y \in N$.

माना कि $y = \sqrt{\log_2 3 \cdot \log_2 12 \cdot \log_2 48 \cdot \log_2 192 + 16} - \log_2 12 \cdot \log_2 48 + 10$ यदि $y \in N$ तो y का मान ज्ञात कीजिए।

Ans. $y = 6$

Sol. Let $y = \sqrt{\log_2 3 \cdot \log_2 12 \cdot \log_2 48 \cdot \log_2 192 + 16} - \log_2 12 \cdot \log_2 48 + 10$

$$y = \sqrt{\log_2 3 \cdot \left(\log_2 2^2 \cdot 3\right) \cdot \left(\log_2 2^4 \cdot 3\right) \cdot \left(\log_2 2^6 \cdot 3\right) + 16} - (\log_2 2^2 \cdot 3) \cdot (\log_2 2^4 \cdot 3) + 10$$

$$y = \sqrt{\log_2 3 \cdot (2 + \log_2 3) \cdot (4 + \log_2 3) \cdot (6 + \log_2 3) + 16} - (2 + \log_2 3) \cdot (4 + \log_2 3) + 10$$

put $\log_2 3 = x$

$$y = \sqrt{x(x+2)(x+4)(x+6)+16} - (x+2)(x+4) + 10$$

$$y = \sqrt{\left(x^2+6x\right)\left(x^2+6x+8\right)+16} - \left(x^2+6x+8\right) + 10$$

put $t = x^2 + 6x$

$$y = \sqrt{t(t+8)+16} - (t+8) + 10$$

$y = t + 4 - t - 8 + 10$

$y = 6$

9. Find 'a' for which the equation $x^2 + (a - b)x + (1 - a - b) = 0$ has two distinct and unequal roots for $\forall b \in R$?

'a' के वह मान ज्ञात कीजिए जिसके लिए समीकरण $x^2 + (a - b)x + (1 - a - b) = 0$ के दो भिन्न–भिन्न तथा असमान मूल हों, जहाँ $\forall b \in R$?

Ans. $a > 1$

Sol. $x^2 + (a - b)x + (1 - a - b) = 0$

$D > 0$

$(a - b)^2 - 4(1 - a - b) > 0$

$a^2 + b^2 - 2ab - 4 + 4a + 4b > 0$

$b^2 + b \cdot (4 - 2a) + (a^2 + 4a - 4) > 0$

$D < 0$ for $\forall b \in R$

$(4 - 2a)^2 - 4 \cdot 1 \cdot (a^2 + 4a - 4) < 0$

$16 + 4a^2 - 16a - 4a^2 - 16a + 16 < 0$

$32 - 32a < 0$

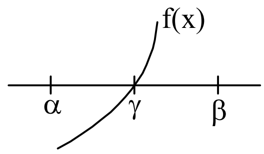
$a > 1$

10. Let a, b, c be real numbers, $a \neq 0$. If α is root of $a^2x^2 + bx + c = 0$. β is the root of $a^2x^2 - bx - c = 0$ and $0 < \alpha < \beta$, then show that the equation $a^2x^2 + 2bx + 2c = 0$ has a root γ that always satisfies $\alpha < \gamma < \beta$.

माना a, b, c वास्तविक संख्याऐं है, जहाँ $a \neq 0$ यदि α समीकरण $a^2x^2 + bx + c = 0$ का एक मूल है तथा β समीकरण $a^2x^2 - bx - c = 0$ का एक मूल जहाँ $0 < \alpha < \beta$, तब प्रदर्शित कीजिए कि समीकरण $a^2x^2 + 2bx + 2c = 0$ का मूल γ है जो संबंध $\alpha < \gamma < \beta$ को हमेशा संतुष्ट करता है।

Ans.

Sol. $\because$ α is root of $a^2x^2 + bx + c = 0$



$a^2\alpha^2 + b\alpha + c = 0$

$b\alpha + c = -a^2\alpha^2$ ...(1)

β is root of $a^2x^2 - bx - c = 0$

$a^2\beta^2 - b\beta - c = 0$

$a^2\beta^2 = b\beta + c$ ...(2)

Let $f(x) = a^2x^2 + 2bx + 2c$

$f(\alpha) = a^2\alpha^2 + 2(b\alpha + c)$

$= a^2\alpha^2 - 2a^2\alpha^2$

$= -a^2\alpha^2 < 0$

$f(\beta) = a^2\beta^2 + 2b\beta + 2c$

$= a^2\beta^2 + 2(b\beta + c)$

$= 3\alpha^2\beta^2 > 0$

and given that

γ is root of $a^2x^2 + 2bx + 2c = 0$

$\therefore$ $\alpha < \gamma < \beta$

11. If α, β are the roots of the equation, $(\tan^2 135°)x^2 - (\text{cosec}10° - \sqrt{3}\sec 10°)x + \tan^2 240° = 0$ then find the quadratic equation whose roots are $(2\alpha + \beta)$ and $(\alpha + 2\beta)$.

यदि α, β समीकरण $(\tan^2 135°)x^2 - (\text{cosec}10° - \sqrt{3}\sec 10°)x + \tan^2 240° = 0$ के मूल है तो वह द्विघात समीकरण ज्ञात कीजिए जिसके मूल $(2\alpha + \beta)$ तथा $(\alpha + 2\beta)$ है।

Ans. $x^2 - 12x + 35 = 0$

Sol. $\left(\tan^2 135^{\circ}\right)x^2 - \left(\text{cosec}10^{\circ} - \sqrt{3}\sec 10^{\circ}\right)x + \tan^2 240^{\circ} = 0 \left\langle \begin{matrix}\alpha \\ \beta\end{matrix}\right.$

$$\alpha + \beta = \frac{\text{cosec}10^{\circ} - \sqrt{3}\sec 10^{\circ}}{\tan^2 135^{\circ}} = 4$$

a new quadratic equation whose roots are : $(2\alpha + \beta)$ and $(\alpha + 2\beta)$

$= (\alpha + \beta + \alpha)$ ad $(\alpha + \beta + \beta)$

$= (4 + \alpha)$ and $(4 + \beta)$

quad. in y : $y = 4 + \alpha$

$y = 4 + x$

put $x = y - 4$

$\left(\tan^2 135^\circ\right)\cdot\left(y-4\right)^2-\left(\operatorname{cosec}10^\circ-\sqrt{3}\sec 10^\circ\right)\cdot\left(y-4\right)+\tan^2 240^\circ=0$

$(y-4)^2-4(y-4)+3=0$

$y^2+16-8y-4y+16+3=0$

$y^2-12y+35=0$

12. Let $x=\frac{1}{3}$ or $x=-15$ satisfies the equation, $\log_8(kx^2+wx+f)=2$. If k, w and f are relatively prime positive integers then find the value of $k+w+f$.

माना $x=\frac{1}{3}$ या $x=-15$ समीकरण $\log_8(kx^2+wx+f)=2$ को संतुष्ट करते है। यदि k, w तथा f सापेक्षतः अभाज्य धनात्मक संख्याऐं हो तो $k+w+f$ का मान ज्ञात कीजिए।

Ans. 96

Sol. Let $x=\frac{1}{3}$ or $x=-15$ satisfies the equation

$\log_8(kx^2+wx+f)=2$

$kx^2+wx+f=64$

$$kx^2+wx+f-64=0\left\langle\begin{matrix}\frac{1}{3}\\-15\end{matrix}\right.$$

sum of roots $\Rightarrow \frac{1}{3}-15=-\frac{w}{k}$

$-\frac{44}{3}=-\frac{w}{k}$

$3w=44k$

$w=\frac{44k}{3}$

Product of roots $\Rightarrow \frac{1}{3}\times(-15)=\frac{f-64}{k}$

$-5k=f-64$

$f=64-5k$

for +ve f; k = 12

and given that k, w and f are relatively prime positive integers (whose HCF = 1)

for $w=\frac{44k}{3}$

k can be; 3,6,9,12

for k = 3, $w=44\times\frac{3}{3}=44$, f = 64 – 15 = 9 (Common factor = 1) for k = 6, $w = 44 \times \frac{6}{3} = 88$, (Common factor = 1,2)

for k = 9, $w = 44 \times \frac{9}{3} = 132$ (Common factor = 1,3)

for k = 12, $w = 44 \times \frac{12}{3} = 176$ (Common factor = 1,4)

$\therefore$ k = 3, w = 44, f = 49

$\Rightarrow$ k + w + f = 3 + 44 + 49 = 96

13. If roots of the equation $x^2 - 10cx - 11d = 0$ are a, b and those of $x^2 - 10ax - 11b = 0$ are c, d, then find the value of a + b + c + d. (a, b, c and d are distinct numbers)

यदि समीकरण $x^2 - 10cx - 11d = 0$ के मूल a, b है तथा $x^2 - 10ax - 11b = 0$ के मूल c, d है तो a + b + c + d का मान बताइये। (a, b, c व d परस्पर भिन्न संख्याऐं है)

Ans. 1210

Sol. $x^2 - 10cx - 11d = 0 \begin{matrix} a \\ b \end{matrix}$ ...(1) $x^2 - 10ax - 11b = 0 \begin{matrix} c \\ d \end{matrix}$ ...(2)

a + b = 10c c + d = 10a

$\Rightarrow$ a + b – c – d = 10c – 10a

b – d = 10c – 10a + c – a

(b – d) = 11(c – a)

$\Rightarrow$ a + b + c + d = 10 (c + a)

from equation (1)

$a^2 - 10ca - 11d = 0$ ...(3)

from equation (2)

$c^2 - 10ca - 11b = 0$ ...(4)

from (3) – (4)

$a^2 - c^2 + 11(b - d) = 0$

(a – c) (a + c) = –11(b – d)

–(c – a) (c + a) = –11·11(c – a)

c + a = 121

$\Rightarrow$ a + b + c + d = 10 × 121 = 1210

COMPREHENSION TYPE QUESTIONS

Paragraph for Question Nos. 14 to 16

Consider the cubic equation

$$x^3 - (1 + \cos\theta + \sin\theta)x^2 + (\cos\theta \sin\theta + \cos\theta + \sin\theta)x - \sin\theta \cdot \cos\theta = 0$$

whose roots are x_1, x_2 and x_3.

तीन घातीय समीकरण

$$x^3 - (1 + \cos\theta + \sin\theta)x^2 + (\cos\theta \sin\theta + \cos\theta + \sin\theta)x - \sin\theta \cdot \cos\theta = 0$$

के मूल x_1, x_2 तथा x_3 है।

14. The value of $x_1^2 + x_2^2 + x_3^2$ equals :

$x_1^2 + x_2^2 + x_3^2$ का मान बराबर है :

(A) $2\cos\theta$ (B) $\sin\theta(\sin\theta + \cos\theta)$ (C) 1 (D) 2

Ans. D

Sol. $S_1 = x_1 + x_2 + x_3 = 1 + \cos\theta + \sin\theta$

$S_2 = \Sigma x_1 \cdot x_2 = (\cos\theta \sin\theta + \cos\theta + \sin\theta)$

$S_3 = x_1 \cdot x_2 \cdot x_3 = \sin\theta \cdot \cos\theta$

$x_1^2 + x_2^2 + x_3^2 = (x_1 + x_2 + x_3)^2 - 2 \cdot \sum x_1 \cdot x_2$

$= (1 + \cos\theta + \sin\theta)^2 - 2 \cdot (\cos\theta \sin\theta + \cos\theta + \sin\theta)$

$= 1 + \cos^2\theta + \sin^2\theta + 2\cos\theta + 2\cos\theta \sin\theta + 2\sin\theta - 2\cos\theta \sin\theta - 2\cos\theta - 2\sin\theta$

$= 1 + 1 = 2$

15. Number of values of θ in $[0, 2\pi]$ for which at least two roots are equal :

θ के अन्तराल $[0, 2\pi]$ में समीकरण के दो मूलों में अधिकतम संभावित अंतर है :

(A) 6 (B) 5 (C) 4 (D) 3

Ans. B

Sol. $x^3 - x^2 - x^2(\cos\theta + \sin\theta) + x(\cos\theta + \sin\theta) + x\cos\theta \sin\theta - \sin\theta \cos\theta = 0$

$x^2 \cdot (x - 1) - x \cdot (\cos\theta + \sin\theta)(x - 1) + (x - 1)\sin\theta \cdot \cos\theta = 0$

$(x-1)\left[x^2 - x(\cos\theta + \mathrm{si}\theta) + \sin\theta \cdot \cos\theta\right] = 0$

$(x - 1)(x - \cos\theta)(x - \sin\theta) = 0$

$x = 1$ or $x = \cos\theta$ or $x = \sin\theta$

At least two roots are equal

$\sin\theta = 1$ or $\cos\theta = 1$ or $\sin\theta = \cos\theta$

in $\theta \in [0, 2\pi]$

$\theta = \dfrac{\pi}{2}$ $\theta = 2\pi, 0$ $\theta = \dfrac{\pi}{4}, \dfrac{5\pi}{4}$

and $\sin\theta = \cos\theta = 1$ (No value of θ)

total values of $\theta = 5$

16. Greatest possible difference between two of the roots if $\theta \in [0, 2\pi]$ is :

यदि θ के अन्तराल $[0, 2\pi]$ में समीकरण के दो मूलों में अधिकतम संभावित अन्तर है :

(A) 2 (B) 1 (C) $\sqrt{2}$ (D) $2\sqrt{2}$

Ans. A

Sol. difference of two roots :

$$d_1 = |\sin\theta - \cos\theta| = \sqrt{2}$$

$$\left.\begin{array}{l} d_2 = |1-\sin\theta| = 2 \\ d_3 = |1-\cos\theta| = 2 \end{array}\right]_{max.}$$

MATCH THE COLUMN TYPE QUESTION(S)

17. Match the following columns :

Column-I	Column-II
(A) If $\alpha, \beta \in (0, \pi)$ and $\alpha \neq \beta$ satisfy the equation $\frac{1-\cos 2\theta}{\sin\theta} = \frac{1}{2}$, then the value of $\tan\left(\frac{\alpha}{2}\right) + \tan\left(\frac{\beta}{2}\right)$ is equal to	(P) 0
(B) If the expression $\frac{x^2+(2m+3)x+(m^2+3)}{\sqrt{x^2+(2m+1)x+m^2+2}}$ is non-negative $\forall\, x \in R$, then the possible values of m can be equal to	(Q) 8
(C) If the parabola $y = 5x^2 + x - 3$ lies above the parabola $y = 2x^2 + 6x - 1$, then integral values of x can be equal to	(R) 1
(D) The number of real solutions of the equation $x^{2\log_x(x+3)} = 16$ is equal to	(S) – 1
	(T) 2

निम्न Columns को सुमेलित कीजिए :

Column-I	Column-II
(A) यदि $\alpha, \beta \in (0, \pi)$ तथा $\alpha \neq \beta$ समीकरण $\frac{1-\cos 2\theta}{\sin\theta} = \frac{1}{2}$ को संतुष्ट करता है तथा $\tan\left(\frac{\alpha}{2}\right) + \tan\left(\frac{\beta}{2}\right)$ का मान बराबर है	(P) 0
(B) यदि व्यंजक $\frac{x^2+(2m+3)x+(m^2+3)}{\sqrt{x^2+(2m+1)x+m^2+2}}$ $\forall\, x \in R$ ऋणात्मक नहीं है, तब m का संभव मान है	(Q) 8
(C) यदि परवलय $y = 5x^2 + x - 3$ परवलय $y = 2x^2 + 6x - 1$ के ऊपर स्थित है, तो x का पूर्णांक मान है	(R) 1
(D) समीकरण $x^{2\log_x(x+3)} = 16$ के वास्तविक हलों की संख्या बराबर है	(S) – 1
	(T) 2

(A) A-P,S; B-Q; C-Q,S; D-P
(B) A-P; B-Q,S; C-Q; D-P,S
(C) A-Q; B-P,S; C-Q,S; D-P
(D) A-P; B-Q,S; C-P,S; D-Q

Ans. (C)

Sol. (A) If $\alpha, \beta \in (0, \pi)$ and $\alpha \neq \beta$

$$\frac{1-\cos 2\theta}{\sin\theta} = \frac{1}{2}$$

$$\frac{2\sin^2\theta}{\sin\theta} = \frac{1}{2}$$

$$\sin\theta = \frac{1}{4}$$

$$\frac{2\tan\frac{\theta}{2}}{1+\tan^2\frac{\theta}{2}} = \frac{1}{4}$$

$$\tan^2\frac{\theta}{2} - 8\tan\frac{\theta}{2} + 1 = 0 \left\langle \begin{matrix}\alpha \\ \beta\end{matrix}\right.$$

$$\tan\left(\frac{\alpha}{2}\right) + \tan\left(\frac{\beta}{2}\right) = 8$$

$A \to Q$

(B) $\dfrac{x^2 + (2m+3)x + (m^2+3)}{\sqrt{x^2 + (2m+1)x + m^2 + 2}} \geq 0$

$x^2 + (2m + 3)x + (m^2 + 3) \geq 0$
$D \leq 0$
$(2m + 3)^2 - 4(m^2 + 3) \leq 0$
$4m^2 + 9 + 12m - 4m^2 - 12 \leq 0$
$12m - 3 \leq 0$

$$m \leq \frac{1}{4}$$

$x^2 + (2m + 1)x + m^2 + 2 > 0$
$D < 0$
$(2m + 1)^2 - 4(m^2 + 2) < 0$
$4m^2 + 1 + 4m - 4m^2 - 8 < 0$
$4m < 7$

$$m < \frac{7}{4}$$

Possible value of $m \in \left(-\infty, \frac{1}{4}\right]$

(C) $5x^2 + x - 3 > 2x^2 + 6x - 1$

$3x^2 - 5x - 2 > 0$

$3x^2 - 6x + x - 2 > 0$

$3x(x-2) + (x-2) > 0$

$(3x+1)(x-2) > 0$

$x \in \left(-\infty, -\frac{1}{3}\right) \cup (2, \infty)$

(D) $x^{2\log_x(x+3)} = 16$

$(x+3)^2 = 16;$ $\quad\quad$ $x > 0, x \neq 1$

$x^2 + 6x - 7 = 0$

$x = 1, -7$

$\therefore$ Number of real solution = 0

$D \to P$

18. Match the following columns :

Column-I	**Column-II**
(A) The expression, $x^{\ln y - \ln z} . y^{\ln z - \ln x} . z^{\ln x - \ln y}$ when simplified reduces to	(P) $\frac{1}{8}$
(B) Let $p = \log_5 \log_5(3)$. If $3^{C+5^{-p}} = 45$, the value of C equals	(Q) $\frac{1}{4}$
(C) The real value of x for which the statement $\log_6 9 - \log_9 27 + \log_8 x = \log_{64} x - \log_6 4$ holds true, is	(R) $\frac{1}{2}$
(D) $\frac{1}{2}\log_{\sqrt{5}} \sin\left(\frac{\pi}{5}\right).\log_{\sqrt{\sin\frac{\pi}{5}}} 5$ simplifies to	(S) 1
	(T) 2

निम्न Columns को सुमेलित कीजिए :

Column-I	**Column-II**
(A) व्यंजक $x^{\ln y - \ln z} . y^{\ln z - \ln x} . z^{\ln x - \ln y}$ का मान होगा	(P) $\frac{1}{8}$
(B) यदि $p = \log_5 \log_5(3)$ तथा $3^{C+5^{-p}} = 45$ तो C का मान होगा	(Q) $\frac{1}{4}$
(C) x के किस मान के लिए समीकरण $\log_6 9 - \log_9 27 + \log_8 x = \log_{64} x - \log_6 4$ संतुष्ट होता है	(R) $\frac{1}{2}$

(D) $\frac{1}{2}\log_{\sqrt{5}} \sin\left(\frac{\pi}{5}\right).\log_{\sqrt{\sin\frac{\pi}{5}}} 5$ का मान होगा (S) 1

(T) 2

(A) A-S; B-T; C-P; D-T

(B) A-T; B-S; C-P; D-T

(C) A-P; B-T; C-T; D-S

(D) A-T; B-P; C-T; D-S

Ans. (A)

Sol. (A) Let $P = x^{\ln y-\ln z}\cdot y^{\ln z-\ln x}\cdot z^{\ln x-\ln y}$

$\log P = (\log y - \log z)\cdot \log x + (\log z - \log x)\cdot \log y + (\log x - \log y)\cdot \log z$

$\log P = \log x\cdot \log y - \log z\cdot \log x + \log y\cdot \log z - \log x\cdot \log y + \log z\cdot \log x - \log y\cdot \log z$

$\log P = 0$

$P = 1$

(B) $P = \log_5 \log_5 (3)$

And $3^{c+5^{-P}} = 45$

$(c + 5^{-P}) \log 3 = \log 45$

$c + 5^{-P} = \log_3 45$

$c = \log_3 45 - 5^{-P}$

$= \log_3 45 - 5^{-\log_5 \log_5 (3)}$

$= \log_3 45 - (\log_5 3)^{-1}$

$= \log_3 \frac{45}{5} = \log_3 9 = 2$

(C) $\log_6 9 - \log_9 27 + \log_8 x = \log_{64} x - \log_6 4$

$\log_6 36 - \frac{3}{2} + \log_8 x - \frac{1}{2}\log_8 x = 0$

$2 - \frac{3}{2} + \frac{1}{2}\log_8 x = 0$

$\frac{1}{2}\log_8 x = -\frac{1}{2}$

$x = \frac{1}{8}$

(D) $\frac{1}{2}\log_{\sqrt{5}} \sin\left(\frac{\pi}{5}\right)\cdot \log_{\sqrt{\sin\frac{\pi}{5}}} 5$

$= \frac{1}{2}\cdot 2\cdot \log_5 \sin\frac{\pi}{5}\cdot 2\log_{\sin\frac{\pi}{5}} 5$

$= 2 \times 1 = 2$

19. Match the following columns :

	Column-I	Column-II
A.	If equation $x^2 + px + q = 0$ $(p, q \in R)$ and $x^3 + 3x^2 + 5x + 3 = 0$ has two common roots then $(p + q)$	P. 1
B.	If $f(x) = 4x^2 + 3x - 7$ and α is common roots of equation $x^2 - 3x + 2 = 0$ and $x^2 + 2x - 3 = 0$ then value of $f(\alpha) =$	Q. 2
C.	Number of real roots of equation $\log_3 (x^2 + 9) + \log_3 (x^4 + 81) = 6$	R. 0
D.	Number of real roots of equation $\log_x(5x - 2) = 3$	S. 5

(A) A-R; B-S; C-Q; D-P
(B) A-S; B-R; C-P; D-Q
(C) A-Q; B-P; C-S; D-R
(D) A-P; B-Q; C-S; D-R

निम्न Columns को सुमेलित कीजिए :

	Column-I	Column-II
A.	यदि समीकरण $x^2 + px + q = 0$ $(p, q \in R)$ तथा $x^3 + 3x^2 + 5x + 3 = 0$ के दो मूल उभयनिष्ठ है, तो $(p + q)$ का मान होगा।	P. 1
B.	यदि $f(x) = 4x^2 + 3x - 7$ तथा α समीकरण $x^2 - 3x + 2 = 0$ तथा $x^2 + 2x - 3 = 0$ का उभयनिष्ठ मूल हो, तो $f(\alpha)$ बराबर है	Q. 2
C.	समीकरण $\log_3 (x^2 + 9) + \log_3 (x^4 + 81) = 6$ के वास्तविक हलों की संख्या है	R. 0
D.	समीकरण $\log_x(5x - 2) = 3$ के वास्तविक हलों की संख्या है	S. 5

(A) A-R; B-S; C-Q; D-P
(B) A-S; B-R; C-P; D-Q
(C) A-Q; B-P; C-S; D-R
(D) A-P; B-Q; C-S; D-R

Ans. B

Sol. (A) $x^2 + px + q = 0, (p, q \in R)$
$x^3 + 3x^2 + 5x + 3 = 0$
$(x + 1)(x^2 + 2x + 3) = 0$
$D < 0$
$\therefore$ roots of x2 + 2x + 3 = 0 will be imaginary and occurs in conjugate pair
$\therefore$ Equation $x^2 + px + q = 0$
and $x^2 + 2x + 3 = 0$ have both common root.
$\therefore\ p = 2, q = 3$
$p + q = 2 + 3 = 5$
$A \rightarrow S$
(B) if $f(x) = 4x^2 + 3x - 7$
And α is common root of equation.
$x^2 - 3x + 2 = 0,$ $\qquad x^2 + 2x - 3 = 0$
$(x - 2)(x - 1) = 0$ $\qquad (x + 3)(x - 1) = 0$
$x = 2, 1$ $\qquad x = -3, 1$
$\Rightarrow \alpha = 1$

$\therefore$ f(α) = f(1) = 4 + 3 – 7 = 0

(C) $\log_3 (x^2 + 9) + \log_3 (x^4 + 81) = 6$

$\log_3 (x^2 + 9)(x^4 + 81) = 6$

$(x^2 + 9)\cdot(x^4 + 81) = 3^6$

$x^6 + 81x^2 + 9x^4 + 729 = 729$

$x^2\cdot[x^4 + 9x^2 + 81] = 0$

$x = 0$

$x^4 + 9x^2 + 81 = 0$

put $x^2 = t$

$t^2 + 9t + 81 = 0$

$\because$ D < 0

$\therefore$ roots are not real.

$\therefore$ Only one root is real.

$C \to P$

(D) $\log_x (5x - 2) = 3;\quad x \neq 1,\qquad 5x - 2 > 0$

$x^3 - 5x + 2 = 0 \qquad x > \dfrac{2}{5}$

$(x - 2)\cdot(x^2 + 2x - 1) = 0$

$x = 2,\ -1-\sqrt{3}, \sqrt{3-1}$

Final $x = 2,\ \sqrt{3} - 1$

$D \to Q$